

Achille Rossi

JUNIOR DEVELOPER · UNDERGRADUATE CS STUDENT

Reggio Emilia - Parma, Italy

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Education

University of Parma

B.S. IN COMPUTER SCIENCE

Parma, Italy

Sep. 2024 - Present

- Best Marks: **Programming Fundamentals A+B, Computer architecture, Algorithms & DS lab., Mobile Development**

ITIS Leonardo da Vinci

HIGH SCHOOL DIPLOMA IN INFORMATION TECHNOLOGY, GRADE: 95/100

Carpi, Italy

Sep. 2019 - Jun. 2024

Skills

Programming Python, C/C++, Java, Node.JS, Database (SQL), Flutter, Kotlin & Android Mobile Development

Web Django with Python, Express with Node.JS, TypeScript, HTML, CSS

Languages Italian, English

Experience

Netribe Ultra

LOW CODE DEVELOPER

Reggio Emilia, Italy

July 2024 - Present

- Implemented an internal company timesheet application for tracking and allocating working hours using the Mendix low-code platform.
- Implemented an automated deployment system in C#.
- Collaborated with clients to design and develop custom business management software.

Dualbit S.R.L.

SCHOOL WORK PLACEMENT – SOFTWARE DEVELOPMENT

Carpi, Italy

Jan. 2024 - Feb. 2024

- Developed a functional software application as part of a school work-placement programme.
- Implemented a web application for managing and tracking the progress of software development projects using Django with Python.

Academic Activity

Match Movie

KOTLIN, ANDROID STUDIO

2026

- Developer a mobile Android application for movie recommendation based on user preferences and movie database.
- Integrated an AI assistant to provide personalized movie suggestions and enhance user experience.
- Integrated a movie database API to fetch real-time information about movies, including ratings, reviews, and trailers.
- Source Code: github.com/akij22/Match-Movie

Code of Algorithms & DS course

C, C++

2026

- Implemented algorithms and data structures in C++ for various problems, including sorting, searching, and graph traversal.

Extracurricular Activity

Agentic Worktree

ELECTRON, TYPESCRIPT

2026

- Implemented a cross-platform desktop application for managing and organizing workspaces and projects
- Integrated different coding agent (Codex, Opencode) to provide code generation and completion features for developers.
- Source Code: github.com/akij22/agentic-worktrees

Pizza Study

Typescript

2026

- Built a full-stack Retrieval-Augmented Generation (RAG) chat application for question answering over study documents using React, Vite, Hono, and TypeScript.
- Implemented document ingestion, sentence-aware chunking, embeddings, and vector similarity search with PostgreSQL, pgvector, and Drizzle ORM.
- Integrated streaming AI responses with the Vercel AI SDK and OpenRouter, including citations from retrieved study materials.
- Developed task management, calendar planning, and study dashboard features with typed APIs and shared Zod contracts.
- Source Code: github.com/akij22/pizza-study

Object Pool Manager

C++

2026

- Developed a header-only, type-aware memory pool that recycles aligned raw storage to reduce repeated heap allocations in latency-sensitive systems.
- Implemented in-place object construction with placement new and RAII ownership through `std::unique_ptr` handles with custom deleters.
- Added pointer validation, deterministic resource recycling, on-demand pool expansion, and runtime allocation statistics.
- Built standard-library and GoogleTest suites covering lifecycle management, automatic expansion, container ownership, invalid pointers, and high-churn workloads.
- Source Code: github.com/akij22/object-pool-manager

Arena Allocator

C++

2026

- Implemented a linear (bump) allocator over a preallocated contiguous memory arena for fast grouped allocations.
- Added alignment-aware allocation, capacity checks with `std::bad_alloc`, usage metrics, and constant-time reset-based reclamation.
- Provided a placement-construction helper for typed objects together with factory and lifecycle management APIs.
- Built Makefile demos and tests covering alignment, capacity exhaustion, arena reuse, object construction, and lifecycle behavior.
- Source Code: github.com/akij22/arena-allocator